

Course Code:B23ADOE06					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
PRINCIPLES OF INTERNET OF THINGS					
Offered by AIDS					
(offered to CE, ECE, EEE & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	List the layers in the modified OSI model for IoT systems.	1	2	2
	b).	State the importance of M2M communication in IoT applications.	1	2	2
	c).	State the role of message communication protocols in connected devices.	2	2	2
	d).	Differentiate between HTTP and HTTPS.	2	2	2
	e).	State the significance of NB-IoT in long-range communication.	3	2	2
	f).	Discuss the need of LoRa in IoT communication.	3	2	2
	g).	State the role of actuators in IoT systems.	4	2	2
	h).	Define On-site processing in IoT.	4	2	2
	i).	List the Components of Vehicular IoT.	5	2	2
	j).	List any two features of Health care IoT.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Write short notes on Real time examples of IoT.	1	2	5
	b).	Explain Machine-to-Machine (M2M) communication and its role in IoT.	1	2	5
		OR			
3.	a).	Explain Modified OSI Model for the IoT/M2M Systems.	1	2	5
	b).	Explain wireless technologies for communication of data from and to devices in IoT.	1	2	5
		UNIT-2			
4.	a).	Explain Web Communication Protocols for Connected devices.	2	2	5
	b).	Discuss message communication protocols for Connected Devices.	2	2	5

		OR			
5.	a).	Explain IP Addressing in IoT.	2	2	5
	b).	Examine Application Layer Protocols in IoT.	2	2	5
		UNIT-3			
6.	a).	Explain Bluetooth, Wi-Fi and Zigbee.	3	2	5
	b).	Describe Identification technologies RFID and NFC.	3	2	5
		OR			
7.	a).	Discuss the IoT Data protocols MQTT and CoAP.	3	2	5
	b).	Explain IoT device management & semantic protocols.	3	2	5
		UNIT-4			
8.	a).	Explain Actuator types in IoT.	4	2	5
	b).	Discuss sensorial deviations in IoT.	4	2	5
		OR			
9.	a).	Explain data formats in IoT and discuss the characteristics of structured and unstructured data.	4	2	5
	b).	Explain the different processing topologies in IoT.	4	2	5
		UNIT-5			
10.	a).	Explain emerging Pillars of IoT.	5	2	5
	b).	Discuss the Challenges Associated with IoT.	5	2	5
		OR			
11.	a).	Apply IoT in the field of Agriculture with an example.	5	3	5
	b).	Apply IoT in the field of Vehicular systems.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23AMOE08					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
IOT BASED SMART SYSTEMS					
Offered by AIML					
(offered to CE, ECE, EEE & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define Internet of Things (IoT).	1	2	2
	b).	Explain about sensors and actuators in IoT?	1	2	2
	c).	Explain the IoT reference model?	2	2	2
	d).	Define Bluetooth Low Energy (BLE) beacons.	2	2	2
	e).	Define RFID and where is it used?	3	2	2
	f).	Define 6LoWPAN in IoT?	3	2	2
	g).	What is Big Data Analytics in IoT?	4	1	2
	h).	Explain the role of Raspberry Pi in IoT applications?	4	2	2
	i).	Define Industrial IoT (IIoT)?	5	2	2
	j).	What is meant by Smart Grid?	5	1	2
5 x 10 =50Marks					
		UNIT-1			
2.	a).	Illustrate the overview and characteristics of the Internet of Things.	1	3	5
	b).	Illustrate technology drivers and business drivers of IoT.	1	3	5
		OR			
3.	a).	Describe the hardware and software requirements for IoT systems.	1	3	5
	b).	Discuss typical IoT applications and their implications.	1	3	5

		UNIT-2			
4.	a).	Illustrate the IoT reference model and architecture with diagram.	2	3	5
	b).	Describe the networking topologies used in IoT.	2	3	5
		OR			
5.	a).	Describe the IoT node structure including sensing, processing, communication, and power components.	2	3	5
	b).	Discuss the role of cloud computing in IoT.	2	3	5
		UNIT-3			
6.	a).	Describe the working of NFC and RFID technologies in IoT applications.	3	2	5
	b).	Illustrate Wi-Fi (IEEE 802.11) and Bluetooth technologies used in IoT.	3	2	5
		OR			
7.	a).	Describe the ZigBee protocol and its features in IoT communication.	3	2	5
	b).	Discuss 6LoWPAN and its importance in IoT networking.	3	2	5
		UNIT-4			
8.	a).	Describe the role of Big Data Analytics in IoT systems.	4	3	5
	b).	Describe the architecture and features of Raspberry Pi for IoT applications.	4	3	5
		OR			
9.	a).	Describe the security and dependability requirements in IoT.	4	3	5
	b).	Discuss Arduino platform and its use in IoT development.	4	3	5
		UNIT-5			
10.	a).	Illustrate Industrial IoT and its applications in manufacturing.	5	3	5
	b).	Describe the Smart City applications using IoT technologies.	5	3	5
		OR			
11.	a).	Describe Home Automation systems using IoT.	5	3	5
	b).	Discuss IoT applications in agriculture and environmental monitoring.	5	3	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23AMOE09					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
COMPUTER NETWORKS					
Offered by AIML					
(offered to CE, ECE, EEE & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Expand OSI.	1	2	1
	b).	Give one example of a mesh topology.	1	2	1
	c).	What is the sliding-window size in Stop-and-Wait ARQ?	2	2	1
	d).	Expand PPP (Point-to-Point Protocol)	2	2	1
	e).	Define hop count in routing.	3	2	1
	f).	What is the IPv6 prefix length for a /6 subnet?	3	2	1
	g).	Which protocol is reliable and connection-oriented?	4	2	1
	h).	State the maximum TCP window size when scaling is disabled.	4	2	1
	i).	Expand URL.	5	2	1
	j).	Name any one HTTP request method	5	2	1
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	With a neat diagram, describe how data flows through all seven OSI layers.	1	3	10
		OR			
3.	a).	Define and differentiate LAN, MAN, WAN.	1	2	5
	b).	Explain star, bus, ring topologies with their advantages and disadvantages.	1	3	5
		UNIT-2			
4.	a).	Distinguish between Go-back-N and Selective-Repeat ARQ protocols.	2	3	5
	b).	A12-bit dataword 110100110011 is to be sent using CRC generator	2	3	5

		polynomial $G(x)=x^4+x+1$. Find the transmitted frame.			
		OR			
5.	a).	Explain p-persistent CSMA and its performance.	2	3	5
	b).	An 11-bit Hamming code 10111011010 is received. Detect and correct the error (if any) and retrieve the original 7-bit data.	2	3	5
		UNIT-3			
6.	a).	A Class C network 192.168.1.0 must be split into 6 sub-nets. (i) Find the new subnet mask (dotted & CIDR). (ii) Hosts per subnet. (iii) First & last valid host of subnet 3.	3	3	5
	b).	Describe the leaky-bucket and token-bucket congestion control algorithms.	3	2	5
		OR			
7.	a).	Explain the differences between distance-vector and link-state routing, detailing their update mechanisms.	3	3	10
		UNIT-4			
8.	a).	Draw the TCP segment format and explain each field.	4	3	5
	b).	Discuss slow-start and congestion-avoidance phases in TCP with a congestion window diagram.	4	3	5
		OR			
9.	a).	Compare features of SCTP with TCP.	4	3	5
	b).	Demonstrate connection termination in TCP using a timing diagram.	4	2	5
		UNIT-5			
10.	a).	Explain SMTP in detail. Give its uses, state strengths and weaknesses	5	2	5
	b).	Differentiate persistent and non-persistent HTTP connections with timing diagrams.	5	2	5
		OR			
11.	a).	Explain in detail about DNS and its frame format.	5	2	5
	b).	Explain FTP architecture highlighting control & data connections.	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23AMOE10					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
SOFTWARE ENGINEERING					
Offered by AIML					
(offered to CE, ECE, EEE & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is meant by the “software problem” in software engineering?	1	1	2
	b).	Distinguish between a software process and a software project.	1	2	2
	c).	State one benefit of validating software requirements early in the requirement process.	2	1	2
	d).	Differentiate functional and non-functional requirements	2	2	2
	e).	What is risk management?	3	1	2
	f).	Name two components typically included in a software project schedule.	3	1	2
	g).	List two key concepts in software design.	4	1	2
	h).	What is the main role of software architecture in software engineering?	4	1	2
	i).	Why is code readability important?	5	2	2
	j).	What is the purpose of unit testing in software development?	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.		Compare the waterfall model with the iterative model in terms of their approach to handling software development phases	1	2	10
		OR			
3.		Discuss the main activities included in software project management process.	1	2	10
		UNIT-2			
4.		In what ways do use cases enhance the process of analyzing and specifying functional requirements during requirement analysis?	2	2	10

		OR			
5.		Describe the process of writing a good SRS. What best practices should be followed to ensure clarity, measurability, and testability?	2	2	10
		UNIT-3			
6.		Briefly describe the steps involved in developing a project monitoring plan for a software project	3	2	10
		OR			
7.		Explain the process of effort estimation for a software project and discuss its impact on project staffing decisions	3	2	10
		UNIT-4			
8.	a).	Describe the component and connector view, and provide an example of how it represents system structure	4	2	5
	b).	Why is verification important during detailed design, and what metrics might be used to support it?	4	2	5
		OR			
9.	a).	Compare function-oriented design and object-oriented design in terms of their approach to software development.	4	2	5
	b).	Illustrate how documenting architecture design contributes to a software system's maintainability and scalability	4	2	5
		UNIT-5			
10.	a).	What is code inspection? Outline three key elements typically examined during a code inspection	5	2	5
	b).	How can metrics be used to assess code quality or the effectiveness of the testing process?	5	2	5
		OR			
11.	a).	Contrast black-box testing with white-box testing, including one advantage of each approach.	5	2	5
	b).	Briefly explain how following coding guidelines such as DRY and KISS contributes to better software quality	5	2	5

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code:B23CEOE07					
SAGIRAMA KRISHNAMRAJUENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODELQUESTIONPAPER					
ALTERNATIVE ENERGY SOURCES					
Offered by CE					
(For AIDS, AIML, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE, IT & ME)					
Time:3 Hrs.			Max.Marks:70M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define renewable energy sources.	1	1	2
	b).	What is meant by a hybrid energy system?	1	1	2
	c).	What is solar radiation?	2	1	2
	d).	List any two applications of solar photovoltaic systems.	2	1	2
	e).	What is wind energy conversion?	3	1	2
	f).	Name any two types of wind turbines.	3	1	2
	g).	What is biogas?	4	1	2
	h).	Define biomass gasification.	4	1	2
	i).	What is Ocean Thermal Energy Conversion (OTEC)?	5	1	2
	j).	What is geothermal energy?	5	1	2
5 x 10 =50Marks					
		UNIT-I			
1.	a)	Explain the need for development of renewable energy sources.	1	2	7
	b)	Explain in detail the scarcity of conventional energy sources in global scenario.	1	2	7
		OR			
2.	a)	Explain hybrid systems in detail.	1	2	7
	b)	Explain the reduction potential of carbon dioxide gas.	1	2	7
		UNIT-II			
3.	a)	Explain solar heating and cooling processes with a neat sketch.	2	2	7
	b)	Explain the measurement and estimation of solar radiation.	2	2	7
		OR			

4.	a)	Explain solar photo voltaic conversion processes	2	2	7
	b)	Explain the applications of solar energy conversion processes.	2	2	7
		UNIT-III			
5.	a)	Derive the expression for power generation from wind turbine using Betz model.	3	2	7
	b)	List and explain various types of winds and factors influencing wind generation.	3	2	7
		OR			
6.	a)	Explain the design aspects of windmill.	3	2	7
	b)	Explain the concepts of energy wheeling and energy banking concepts.	3	2	7
		UNIT-IV			
7.	a)	Explain how biomass is converted by gasification process.	4	2	7
	b)	Explain the production of biogas and discuss its composition, advantages, and applications.	4	2	7
		OR			
8.	a)	Explain how urban waste is collected and converted into energy.	4	2	7
	b)	Explain how biomass is converted by pyrolysis and liquefaction processes.	4	2	7
		UNIT-V			
9.	a)	Explain the theory and working principle of ocean thermal energy conversion(OTEC) system.	5	2	7
	b)	Explain with a neat sketch, the operation of geothermal power plant.	5	2	7
		OR			
10.	a)	Explain with a neat sketch, the working and operation of any turbine used for hydropower generation.	5	2	7
	b)	Explain the tidal energy conversion process with a neat sketch	5	2	7

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23CEOEO8					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
WATER SUPPLY SYSTEMS					
Offered by CE					
(For AIDS, AIML, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE, IT & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define water demand.	1	1	2
	b).	List any two factors affecting water consumption.	1	1	2
	c).	What are surface water sources?	2	1	2
	d).	Define groundwater recharge.	2	1	2
	e).	What is potable water?	3	1	2
	f).	Define dual water supply system.	3	1	2
	g).	What is water distribution system?	4	1	2
	h).	List any two methods of water distribution.	4	1	2
	i).	Define industrial wastewater.	5	1	2
	j).	What are effluent standards?	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain various types of water demands and factors affecting water consumption.	1	2	10
		OR			
3.	a).	Illustrate methods used for forecasting population and estimating water demand.	1	3	10
		UNIT-2			
4.	a).	Describe different sources of water and methods of water collection.	2	2	10
		OR			
5.	a).	Explain techniques of desalination and groundwater recharge with neat	2	2	10

		sketches.			
		UNIT-3			
6.	a).	Explain potable and non-potable water concepts and dual water supply systems.	3	2	10
		OR			
7.	a).	Discuss water-related diseases and their prevention methods.	3	2	10
		UNIT-4			
8.	a).	Explain different types of water distribution systems and their components.	4	2	10
		OR			
9.	a).	Apply principles to design a simple water distribution system for a small community.	4	3	10
		UNIT-5			
10.	a).	Describe industrial water requirements and characteristics of wastewater.	5	2	10
		OR			
11.	a).	Explain standards for disposal of industrial effluents and their environmental impact.	5	2	10

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code:B23CBOE05					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
PRINCIPLES OF OPERATING SYSTEMS					
offered by CSBS					
(for CE, ECE, EEE & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	List the Goals of Operating Systems.	1	2	2
	b).	List any two open source Operating Systems.	1	2	2
	c).	List the states of process during its life cycle.	2	2	2
	d).	Why is Interprocess Communication required?	2	2	2
	e).	What is the race condition?	3	1	2
	f).	List the classical synchronization problems..	3	2	2
	g).	What is swapping?	4	1	2
	h).	What is Demand Paging?	4	1	2
	i).	List Different file operations.	5	2	2
	j).	What is an Access matrix?	5	1	2
		UNIT-1	CO	KL	M
2.	a).	Define the operating system and list the basic functions provided by the operating system.	1	2	5
	b).	What is a system call? Explain with its types.	1	2	5
		OR			
3.	a).	List and Explain types of Operating Systems.	1	2	5
	b).	Explain about operating system structure in detail.	1	2	5
		UNIT-2			
4.	a).	What is process scheduling? Explain different types of schedulers?	2	2	5

	b).	Construct the Gantt chart for a) Shortest Remaining time first for the following, <table><tr><td>Process</td><td>P1</td><td>P2</td><td>P3</td><td>P4</td><td>P5</td></tr><tr><td>Arrival time</td><td>0</td><td>0</td><td>2</td><td>1</td><td>3</td></tr><tr><td>Burst time in ms</td><td>10</td><td>6</td><td>2</td><td>8</td><td>5</td></tr></table>	Process	P1	P2	P3	P4	P5	Arrival time	0	0	2	1	3	Burst time in ms	10	6	2	8	5	2	3	5
Process	P1	P2	P3	P4	P5																		
Arrival time	0	0	2	1	3																		
Burst time in ms	10	6	2	8	5																		
		OR																					
5.	a).	Compare single-threaded and multi-threaded models with examples.	2	2	5																		
	b).	Explain FIFO and Round Robin CPU scheduling algorithms. Why do we need them?	2	2	5																		
		UNIT-3																					
6.	a).	Explain about deadlock conditions and Banker's algorithm in detail.	3	2	10																		
		OR																					
7.	a).	How does Peterson's Solution ensure mutual exclusion?	3	2	5																		
	b).	State Readers Writers problem and give solution using semaphore.	3	2	5																		
		UNIT-4																					
8.	a).	Explain how paging supports virtual memory. A neat diagram and explain how logical address is translated into physical address.	4	2	10																		
		OR																					
9.	a).	Explain LRU, FIFO page replacement algorithms for the following example reference 1,3,4,0,5,3,2,1,0,4,5,2,1,3,2,5 and frame size=3.	4	3	5																		
	b).	Compare Different Disk Scheduling Algorithms.	4	2	5																		
		UNIT-5																					
10.	a).	Explain various file allocation methods in detail.	5	2	5																		
	b).	Explain Protection Mechanism illustrating use of protection domain and access control list.	5	2	5																		
		OR																					
11.	a).	Explain about the free space management in the file system.	5	2	5																		
	b).	Write a short note on Revocation of access rights.	5	2	5																		

CO-COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23CSOE05					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
PRINCIPLES OF SOFTWARE ENGINEERING					
offered by CSE					
(Offered to CE, ECE, EEE & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is meant by the “software problem” in software engineering?	1	1	2
	b).	Distinguish between a software process and a software project.	1	2	2
	c).	State one benefit of validating software requirements early in the requirement process.	2	1	2
	d).	Differentiate functional and non-functional requirements	2	2	2
	e).	What is risk management?	3	1	2
	f).	Name two components typically included in a software project schedule.	3	1	2
	g).	List two key concepts in software design.	4	1	2
	h).	What is the main role of software architecture in software engineering?	4	1	2
	i).	Why is code readability important?	5	2	2
	j).	What is the purpose of unit testing in software development?	5	1	2
		UNIT-1			
2.		Compare the waterfall model with the iterative model in terms of their approach to handling software development phases	1	2	10
		OR			
3.		Discuss the main activities included in software project management process.	1	2	10
		UNIT-2			
4.		In what ways do use cases enhance the process of analyzing and specifying functional requirements during requirement analysis?	2	2	10

		OR			
5.		Describe the process of writing a good SRS. What best practices should be followed to ensure clarity, measurability, and testability?	2	2	10
		UNIT-3			
6.		Briefly describe the steps involved in developing a project monitoring plan for a software project	3	2	10
		OR			
7.		Explain the process of effort estimation for a software project and discuss its impact on project staffing decisions	3	2	10
		UNIT-4			
8.	a).	Describe the component and connector view, and provide an example of how it represents system structure	4	2	5
	b).	Why is verification important during detailed design, and what metrics might be used to support it?	4	2	5
		OR			
9.	a).	Compare function-oriented design and object-oriented design in terms of their approach to software development.	4	2	5
	b).	Illustrate how documenting architecture design contributes to a software system's maintainability and scalability	4	2	5
		UNIT-5			
10.	a).	What is code inspection? Outline three key elements typically examined during a code inspection	5	2	5
	b).	How can metrics be used to assess code quality or the effectiveness of the testing process?	5	2	5
		OR			
11.	a).	Contrast black-box testing with white-box testing, including one advantage of each approach.	5	2	5
	b).	Briefly explain how following coding guidelines such as DRY and KISS contributes to better software quality	5	2	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23CSOE06					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
COMPUTER NETWORKS					
offered by CSE					
(Offered to CE, ECE, EEE & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Expand OSI.	1	2	1
	b).	Give one example of a mesh topology.	1	2	1
	c).	What is the sliding-window size in Stop-and-Wait ARQ?	2	2	1
	d).	Expand PPP (Point-to-Point Protocol)	2	2	1
	e).	Define hop count in routing.	3	2	1
	f).	What is the IPv6 prefix length for a /6 subnet?	3	2	1
	g).	Which protocol is reliable and connection-oriented?	4	2	1
	h).	State the maximum TCP window size when scaling is disabled.	4	2	1
	i).	Expand URL.	5	2	1
	j).	Name any one HTTP request method	5	2	1
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	With a neat diagram, describe how data flows through all seven OSI layers.	1	3	10
		OR			
3.	a).	Define and differentiate LAN, MAN, WAN.	1	2	5
	b).	Explain star, bus, ring topologies with their advantages and disadvantages.	1	3	5
		UNIT-2			
4.	a).	Distinguish between Go-back-N and Selective-Repeat ARQ protocols.	2	3	5
	b).	A 12-bit dataword 110100110011 is to be sent using CRC generator	2	3	5

		polynomial $G(x)=x^4+x+1$. Find the transmitted frame.			
		OR			
5.	a).	Explain p-persistent CSMA and its performance.	2	3	5
	b).	An 11-bit Hamming code 10111011010 is received. Detect and correct the error (if any) and retrieve the original 7-bit data.	2	3	5
		UNIT-3			
6.	a).	A Class C network 192.168.1.0 must be split into 6 sub-nets. (i) Find the new subnet mask (dotted & CIDR). (ii) Hosts per subnet. (iii) First & last valid host of subnet 3.	3	3	5
	b).	Describe the leaky-bucket and token-bucket congestion control algorithms.	3	2	5
		OR			
7.	a).	Explain the differences between distance-vector and link-state routing, detailing their update mechanisms.	3	3	10
		UNIT-4			
8.	a).	Draw the TCP segment format and explain each field.	4	3	5
	b).	Discuss slow-start and congestion-avoidance phases in TCP with a congestion window diagram.	4	3	5
		OR			
9.	a).	Compare features of SCTP with TCP.	4	3	5
	b).	Demonstrate connection termination in TCP using a timing diagram.	4	2	5
		UNIT-5			
10.	a).	Explain SMTP in detail. Give its uses, state strengths and weaknesses	5	2	5
	b).	Differentiate persistent and non-persistent HTTP connections with timing diagrams.	5	2	5
		OR			
11.	a).	Explain in detail about DNS and its frame format.	5	2	5
	b).	Explain FTP architecture highlighting control & data connections.	5	2	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23QTOE01					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
QUANTUM SCIENCE AND TECHNOLOGY					
Offered by AIDS, AIML, CIC, CSE, CSBS, IT					
(offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE, IT & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define the photoelectric effect and explain its significance in the development of quantum mechanics.	1	2	2
	b).	State the De Broglie hypothesis and write the expression for the de Broglie wavelength.	1	2	2
	c).	Represent a qubit using the Bloch sphere and explain its geometric interpretation.	2	2	2
	d).	Define Dirac notation (bra-ket) and express a general qubit state using it.	2	2	2
	e).	Explain the concept of quantum parallelism and its role in quantum algorithms.	3	2	2
	f).	What is the Deutsch algorithm, and what problem does it solve?	3	2	2
	g).	Outline the basic steps of the BB84 protocol for quantum key distribution.	4	2	2
	h).	Define quantum teleportation and state its key requirement.	4	2	2
	i).	Define quantum sensors and give one example of their application.	5	2	2
	j).	What does NISQ era stand for, and what are its main challenges?	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the historical significance of the photoelectric effect and Compton scattering in the development of quantum mechanics.	1	3	5
	b).	State and prove the Heisenberg uncertainty principle for position and momentum. Provide an example to illustrate its implications.	1	3	5
		OR			
3.	a).	Describe the concept of wave-particle duality and the de Broglie hypothesis. How does it apply to both light and matter?	1	2	5

	b).	Explain the role of operators in quantum mechanics. How are observables like position, momentum, and energy represented mathematically?	1	2	5
		UNIT-2			
4.	a).	Describe the Bloch sphere representation of a qubit. How does it differ from the classical bit representation?	2	2	5
	b).	Explain the quantum teleportation protocol with a step-by-step analysis of the circuit and measurements involved.	2	2	5
		OR			
5.	a).	Define Dirac notation (bra-ket) and explain how it is used to represent quantum states and operators.	2	2	5
	b).	Discuss the no-cloning theorem and its significance in quantum information theory. Provide a proof outline.	2	2	5
		UNIT-3			
6.	a).	Outline the steps of the Deutsch-Jozsa algorithm and explain how it demonstrates quantum advantage over classical algorithms.	3	3	5
	b).	Explain the principles of quantum error correction using the bit-flip and phase-flip codes. How does Shor's 9-qubit code address both types of errors?	3	2	5
		OR			
7.	a).	Explain the concept of quantum parallelism and how it is utilized in quantum algorithms.	3	2	5
	b).	Compare and contrast classical and quantum oracles. Provide an example of an oracle used in Grover's algorithm.	3	2	5
		UNIT-4			
8.	a).	Explain the concept of quantum dense coding and how it enables the transmission of two classical bits using a single qubit.	4	2	5
	b).	Describe the working of a quantum repeater and its role in enabling long-distance quantum communication.	4	3	5
		OR			
9.	a).	Describe the quantum teleportation protocol with a focus on the role of entanglement and classical communication.	4	2	5
	b).	Discuss the challenges in implementing quantum networks, including loss, decoherence, and noise. How can entanglement swapping mitigate some of these issues?	4	3	5

		UNIT-5			
10.	a).	Define quantum supremacy and discuss the challenges and limitations of the NISQ (Noisy Intermediate-Scale Quantum) era.	5	3	5
	b).	Explain the working of NV centers in diamonds for quantum sensing applications. What are their advantages over traditional sensors?	5	3	5
		OR			
11.	a).	Describe the global initiatives in quantum technologies, including those by IBM, Google, and India's National Quantum Mission (NQM).	5	2	5
	b).	Explain the concept of topological insulators and their potential applications in quantum computing and materials science.	5	3	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code: B23CIOE07					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
INTRODUCTION TO CYBER SECURITY					
offered by CIC					
(Offered to CE, ECE, EEE & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Explain the term cybercrime	1	2	2
	b).	Explain the relationship between cybercrime and information security.	1	2	2
	c).	Explain how phishing attacks are carried out	2	2	2
	d).	Differentiate between virus and worm	2	2	2
	e).	Explain the purpose of eDiscovery tools in collecting digital evidence.	3	2	2
	f).	Explain the importance of evidence preservation during cybercrime investigation.	3	2	2
	g).	Explain the purpose of computer forensics software tools used in digital investigations.	4	2	2
	h).	Explain how fingerprint is used in digital forensic investigations.	4	2	2
	i).	Describe the main objectives of the Information Technology (IT) Act in India.	5	2	2
	j).	Explain the role of digital signatures in cyber law.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the concept of cybercrime, its origins, and the role of cybercriminals in cyber-attacks.	1	2	5
	b).	Describe different types and classifications of cybercrime with suitable examples.	1	2	5
		OR			
3.	a).	Explain the security challenges posed by mobile devices and discuss common attacks on mobile or cell phones	1	2	5
	b).	Describe the concept of cyberstalking and its impact on victims.	1	2	5

		UNIT-2			
4.	a).	Analyze different types of malware such as key loggers, spyware, viruses, and worms	2	4	5
	b).	Analyze foot printing, social engineering, and identity theft techniques used by attackers to gather sensitive information.	2	4	5
		OR			
5.	a).	Analyze the working of proxy servers and anonymizers and explain how attackers use them to hide their identity.	2	4	5
	b).	Analyze DoS and DDoS attacks, explaining their working principles	2	4	5
		UNIT-3			
6.	a).	Analyze the process of e-mail investigation, e-mail tracking, and e-mail recovery in solving cyber crimes	3	4	5
	b).	Explain the role of encryption and decryption methods in cyber security and investigations.	3	2	5
		OR			
7.	a).	Analyze the techniques used in recovering deleted evidence and password cracking during cybercrime investigations.	3	4	10
		UNIT-4			
8.	a).	Explain the process of computer forensic investigation and its importance in cyber crime cases.	4	2	5
	b).	Describe the steps involved in preparing for a computer investigation.	4	2	5
		OR			
9.	a).	Apply the concepts of Windows and Linux system forensics in identifying and collecting digital evidence.	4	3	5
	b).	Explain the role of network forensics, e-mail investigations, and mobile device forensics in solving cyber crimes	4	2	5
		UNIT-5			
10.	a).	Describe the major provisions and features of the Indian Information Technology Act.	5	2	5
	b).	Explain the challenges faced by Indian law in controlling cybercrime and the need for stronger cyber regulations.	5	2	5
		OR			
11.	a).	Apply cyber law provisions to explain punishments for different types of cybercrimes in India.	5	3	10

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks



SRKR
ENGINEERING COLLEGE
AUTONOMOUS

Course Code: B23CIOE08					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
INTRODUCTION TO BLOCKCHAIN TECHNOLOGY					
offered by CIC					
(Offered to CE, ECE, EEE & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Explain the origin of Blockchain.	1	2	2
	b).	Explain different types of Blockchain.	1	2	2
	c).	Explain public Blockchain.	2	2	2
	d).	Explain different types of Oracles	2	2	2
	e).	Explain Smart Contract.	3	2	2
	f).	Discuss Consortium Blockchain system.	3	2	2
	g).	Explain Security Aspects in Bitcoin	4	2	2
	h).	Explain Limitations of Blockchain	4	2	2
	i).	Discuss How to Use Python Online Editor	5	2	2
	j).	Explain the Chain codes.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain the Components of Blockchain.	1	2	5
	b).	Identify the importance of Consensus Protocol in the Blockchain.	1	2	5
		OR			
3.	a).	Explain Decentralization and Distribution in the Blockchain.	1	2	5
	b).	Explain different types of Blockchains.	1	2	5
		UNIT-2			
4.	a).	Explain Public Blockchain system.	2	2	5
	b).	Discuss different Characteristics of a Smart Contract	2	2	5
		OR			
5.	a).	Explain the Ethereum Blockchain system.	2	2	5

	b).	Explain Smart Contracts in Ethereum.	2	2	5
		UNIT-3	3	2	5
6.	a).	Explain Key Characteristics of Private Blockchain.	3	2	5
	b).	Discuss Smart Contract in Private Environment.			
		OR			
7.	a).	Explain Characteristics of Consortium Blockchain	3	2	5
	b).	Discuss the Need of Consortium Blockchain	3	2	5
		UNIT-4			
8.	a).	Explain the Security and Privacy Challenges of Blockchain.	4	2	5
	b).	Illustrate how Blockchain can be used in Banking and Finance.	4	3	5
		OR			
9.	a).	Discuss the process of Safeguarding Blockchain Smart Contract.	4	2	5
	b).	Illustrate how Blockchain can be used in Internet of Things	4	3	5
		UNIT-5			
10.	a).	Illustrate how Python Packages can be used for Blockchain.	5	3	5
	b).	Illustrate how Blockchain can be used in Retail Sector.	5	3	5
		OR			
11.	a).	Explain different Components of Hyperledger Fabric Network	5	2	5
	b).	Illustrate how Blockchain can be used in Health Care.	5	3	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code:B23ECO06					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
PRINCIPLES OF CELLULAR & MOBILE COMMUNICATIONS					
Offered by ECE					
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, EEE, IT &ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer ONE Question from EACH UNIT					
All questions carry equal marks					
Assume suitable data if necessary					
10 X 2=20 Marks					
			CO	KL	M
1	a).	What are the limitations of conventional wireless systems?	1	2	2
	b).	Define frequency reuse ratio.	1	3	2
	c).	Define co-channel interference (CCI).	2	2	2
	d).	What happens when transmit power is reduced?	2	3	2
	e).	Define channel borrowing.	3	2	2
	f).	What is overlaid cells.	3	2	2
	g).	Write a short note about power difference handoffs	4	2	2
	h).	Discuss about call drop rates.	4	3	2
	i).	What are GSM channels?	5	2	2
	j).	What is the main advantage of CDMA over FDMA/TDMA?	5	2	2
5 x 10 = 50 Marks					
		UNIT-I			
2.	a).	Explain about consideration of the components of Cellular system	1	3	5
	b).	Compare 1G, 2G, 3G, and 4G cellular systems in terms of technology, services, and data rates.	1	3	5
		OR			
3.	a).	Derive the co-channel interference reduction factor for seven cell reuse pattern	1	2	5
	b).	Discuss about Trucking and GOS.	1	3	5
		UNIT-II			
4.	a).	Explain the measurement of real time Co-Channel interference.	2	3	5
	b).	With neat sketch explain the effect of Non-co channel interference-adjacent channel interference.	2	3	5

		OR			
5.	a).	Explain co-channel interference (CCI) and derive the expression for Signal-to-Interference Ratio (C/I)..	2	3	5
	b).	Discuss the design of Antenna System with suitable examples.	2	2	5
		UNIT-III			
6.	a).	What is the importance of frequency management chart?	3	3	5
	b).	Discuss about fixed channel and non-fixed channel assignment	3	2	5
		OR			
7.	a).	Explain about channel sharing and borrowing.	3	3	5
	b).	Explain in detail about channel assignments to cell sites and mobile units.	3	3	5
		UNIT-IV			
8.	a).	Explain about handoff initiation and types of handoff.	4	2	5
	b).	Describe the effects of Intersystem and Intrasytem Handoffs.	4	3	5
		OR			
9.	a).	Write about forced handoff and delayed handoff mechanisms in detail.	4	2	5
	b).	Compare soft handoff and hard handoff in detail.	4	3	5
		UNIT-V			
10.	a).	Explain the GSM architecture with a neat diagram, including BSS, NSS, and OSS.	5	3	5
	b).	Compare FDMA, TDMA, and CDMA in terms of capacity, efficiency, and complexity.	5	2	5
		OR			
11.	a).	Explain the concept of MIMO systems and their advantages in wireless communication.	5	2	5
	b).	Discuss the technological advancements in 5G compared to 4G	5	3	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code:B23ECOE07					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
IMAGE PROCESSING					
Offered by ECE					
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, EEE, IT &ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer ONE Question from EACH UNIT					
All questions carry equal marks					
Assume suitable data if necessary					
10 X 2=20 Marks					
			CO	KL	M
1	a).	What is the origin of digital image processing?	1	2	2
	b).	Explain the concept of basic relationships between pixels in an image.	1	1	2
	c).	What is histogram processing, and why is it important in image enhancement?	2	2	2
	d).	What is intensity transformation in image processing?	2	1	2
	e).	What is coding redundancy in image compression?	3	1	2
	f).	What is irrelevant information in image compression?	3	2	2
	g).	Define the terms “opening” and “closing” in morphological processing.	4	3	2
	h).	What is edge detection in image processing?	4	1	2
	i).	Define the HSI color model and explain its components.	4	2	2
	j).	What is the RGB color model?	4	1	2
5 x 10 = 50 Marks					
		UNIT-I			
2.		Describe the fundamental steps involved in digital image processing and their significance.	1	2	10
		OR			
3.		How do the concepts of image sampling and quantization affect the quality of a digital image? Explain with examples.	1	4	10
		UNIT-II			
4.		Describe the basic intensity transformations used in image enhancement. Provide examples and explain their applications.	2	2	10

		OR			
5.		With the help of a block diagram explain basic steps involved in frequency domain filtering.	2	2	10
		UNIT-III			
6.		What are the derivative operators useful in image segmentation? Explain	3	2	5
		OR			
7.	a).	Explain about wavelet-based image processing.	3	2	5
	b).	Discuss the main steps involved in Continuous Wavelet Transform.	3	3	5
		UNIT-IV			
8.	a).	Explain about lossless predictive coding.	4	3	5
	b).	Explain Arithmetic encoding with an example.	4	3	5
		OR			
9.	a).	Explain about image compression models.	4	3	5
	b).	Discuss about the lossy predictive coding	4	2	5
		UNIT-V			
10.	a).	Discuss inverse filtering and how we can improve its performance	4	3	5
	b).	Explain the process of image restoration in presence of noise.	4	2	5
		OR			
11.	a).	Explain the process of image restoration in presence of noise.	4	3	5
	b).	Describe the Hit or miss transformation algorithm in detail.	4	3	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23ECOE08					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
FUNDAMENTALS OF SATELLITE COMMUNICATIONS					
Offered by ECE					
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, EEE, IT &ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer ONE Question from EACH UNIT					
All questions carry equal marks					
Assume suitable data if necessary					
10 X 2=20 Marks					
			CO	KL	M
1	a).	What are the basic components of a satellite communication system?	1	2	2
	b).	State Kepler's laws of planetary motion relevant to satellite communication.	1	1	2
	c).	What is the function of the telemetry, tracking and command (TT&C) subsystem?	2	2	2
	d).	List the major subsystems present in a communication satellite.	2	1	2
	e).	Define C/N ratio and explain its significance in satellite communication.	3	2	2
	f).	What is meant by G/T ratio in satellite link design?	3	1	2
	g).	Compare FDMA and TDMA techniques in satellite communication.	4	2	2
	h).	What is spread spectrum in CDMA?	4	1	2
	i).	What is Global Navigation Satellite System and name any two GNSS constellations?	5	1	2
	j).	Explain the basic principle of GPS positioning.	5	2	2
5 x 10 = 50 Marks					
		UNIT-I			
2.	a).	Explain the three Kepler's laws of planetary motion.	1	2	10
	b).	Explain about various orbital parameters used in orbital analysis of a satellite.	1	1	5
		OR			
3.	a).	Derive the suitable equations for look angles and the range for geostationary satellites.	1	2	5
	b).	Give the condition for satellite visibility criterion.	1	2	5
		UNIT-II			
4.	a).	Examine how the attitude and orbit control system (AOCS) is achieved	2	2	5

		through spin stabilization systems? Give necessary diagrams.			
	b).	Explain the reason for inter-modulation noise originating in a satellite link.	2	2	5
		OR			
5.	a).	Explain TT and C system in detail.	2	2	5
	b).	Express equation for the downlink C/N ratio for the satellite.	2	2	5
		UNIT-III			
6.	a).	Calculate the carrier-to-noise ratio for the combined uplink and downlink communication.	3	3	5
	b).	State how intermodulation noise originates in a satellite link and describe how it is reduced?	3	3	5
		OR			
7.	a).	Explain the effect of rain in both uplink and downlink communication.	3	3	5
	b).	Summarize the sources of noise in satellite communication. What is the importance of noise temperature in link design?	3	3	5
		UNIT-IV			
8.	a).	Illustrate the features of various multiple access schemes deployed for satellite access and compare it.	4	2	5
	b).	Determine the limitations of CDMA.	4	2	5
		OR			
9.	a).	Explain direct sequence spread spectrum communication in detail.	4	2	5
	b).	Write a short note about the SPADE system.	4	2	5
		UNIT-V			
10.	a).	State and explain the working principle behind GPS?	5	3	5
	b).	Explain how receiver position in 3D is determined in GPS with neat diagrams.	5	3	5
		OR			
11.	a).	Explain the function of different segments in the operation of GPS in detail with necessary diagrams.	5	3	5
	b).	Compare GPS, GLONASS and GALILEO in terms of orbit configuration and services.	5	3	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23EEOE07					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
INTRODUCTION TO DRONE TECHNOLOGY					
Offered by EEE					
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, IT &ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	List any two important milestones in the history of drone development.	1	2	2
	b).	State any two Post-flight checklist points for safe drone operation.	1	2	2
	c).	List any two roles of an Electronic Speed Controller (ESC).	2	2	2
	d).	State the purpose of a propeller guard in a quadcopter.	2	2	2
	e).	List the basic forces acting on a quadcopter during flight.	3	2	2
	f).	Mention any two types of propulsion systems used in multirotor drones.	3	2	2
	g).	Justify the need for sensors on a drone?	4	2	2
	h).	Explain the function of a magnetometer in drone systems?	4	2	2
	i).	List any three main components of the Pluto Drone ecosystem	5	2	2
	j).	Mention one feature found in Developer Mode of the Pluto drone app.	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Classify types of drones based on structure and explain in detail multi-rotor drone and Fixed Wing Hybrid VTOL drone with a neat diagrams.	1	3	5
	b).	List out the safety mandates and regulations for drone operations in India.	1	3	5
		OR			
3.	a).	Classify types of drones based on maximum all-up weight including payload.	1	3	5
	b).	List and explain the key points of a pre-flight safety checklist and In-Flight Check list while operating Drones.	1	3	5
		UNIT-2			
4.	a).	Draw and explain the general block diagram of a multirotor drone and	2	3	5

		describe the role of each block.			
	b).	Illustrate the function and working of a Flight Controller.	2	3	5
		OR			
5.	a).	Explain the working of a brushless DC (BLDC) motor.	2	3	5
	b).	Explain the construction and types of drone frames and what kind of materials are used and why?	2	3	5
		UNIT-3			
6.	a).	Discuss in detail the four fundamental movements and controls of a quadcopter (Roll, Pitch, Yaw, Throttle).	3	3	5
	b).	Illustrate and explain the propulsion system of multirotor drones and identify the critical components and their functions?	3	3	5
		OR			
7.	a).	Explain the three principal axes of rotation and how each relates to drone movement (Pitch, Roll, and Yaw).	3	3	5
	b).	Analyze the torque or moments acting on a quadcopter and how the flight controller compensate for them?	3	3	5
		UNIT-4			
8.	a).	Compare the functions of an accelerometer sensor and a barometer sensor in drone flight control.	4	3	5
	b).	Analyze the importance of proximity sensors in obstacle avoidance for drones.	4	3	5
		OR			
9.	a).	Illustrate how GPS sensors enhance autonomous drone navigation with an example.	4	3	5
	b).	Explain the working and importance of drone indicators.	4	3	5
		UNIT-5			
10.	a).	List and explain the key components in the Pluto Drone ecosystem and their functions include a block diagram.	5	3	5
	b).	Explain the process of connecting the Pluto Drone to the mobile app and how the app interface help in control and calibration?	5	3	5
		OR			
11.	a).	Illustrate the step-by-step procedure to build a Pluto Drone from unboxing to first flight.	5	3	5
	b).	Explain the control settings, trim settings, and developer mode options in the Pluto Drone App. How do they enhance flying?	5	3	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks



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Course Code: B23EEOE08					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
ENERGY MANAGEMENT AND AUDIT					
Offered by EEE					
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, IT &ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	List out various types of energy available	1	2	2
	b).	What is primary energy? Give one example.	1	2	2
	c).	Mention any two qualities of a successful energy manager.	2	2	2
	d).	Name two principles of energy management.	2	2	2
	e).	What is waste heat?	3	2	2
	f).	State the function of an air preheater.	3	2	2
	g).	Define power factor.	4	2	2
	h).	Define Life Cycle Cost (LCC).	4	2	2
	i).	Define cost index in energy management.	5	2	2
	j).	What is the purpose of a pie chart in energy auditing?	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Write a detailed note on renewable and non-renewable sources of energy.	1	3	5
	b).	Review the energy scenario in India brief	1	3	5
		OR			
3.	a).	Explain the classification of energy resources with suitable examples.	1	3	5
	b).	Discuss various forms of energy. Explain potential and kinetic energy with suitable examples.	1	3	5
		UNIT-2			
4.	a).	Elaborate the functions and qualities required in an energy manager.	2	3	5
	b).	Explain the role of energy audit in an energy management programme.	2	3	5

		OR			
5.	a).	Describe the necessary steps involved in implementing an effective energy management programme.	2	3	5
	b).	Discuss the principles of energy management.	2	3	5
		UNIT-3			
6.	a).	Describe different types of heat exchangers used in waste heat recovery systems.	3	3	5
	b).	Explain the concept of waste heat and its importance in industrial energy management.	3	3	5
		OR			
7.	a).	Explain the role of air preheaters and economizers in steam systems	3	3	5
	b).	Write a detailed note on steam generation, distribution, and utilization methodology.	3	3	5
		UNIT-4			
8.	a).	Discuss Equivalent Annual Worth (EAW) and how it aids in comparing investment alternatives	4	3	5
	b).	What are energy-efficient motors? Explain types, selection criteria, and economic impact.	4	3	5
		OR			
9.	a).	Discuss the causes and corrective methods of low power factor. Also, explain benefits of PF improvement.	4	3	5
	b).	Describe methods for achieving energy-efficient lighting in commercial and industrial buildings.	4	3	5
		UNIT-5			
10.	a).	Explain the objectives and importance of energy audits in industries.	5	3	5
	b).	Describe the use and significance of Sankey diagrams in energy auditing.	5	3	5
		OR			
11.	a).	What are energy conservation schemes? Describe with examples from industrial or commercial sectors.	5	3	5
	b).	Explain the different types of energy audits and highlight their differences.	5	3	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23ITOE04					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
PRINCIPLES OF OPERATING SYSTEMS					
offered by IT					
(Offered to CE, ECE, EEE & ME)					
Time: 3 Hrs.			Max. Marks: 70M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Define Operating system.	1	2	2
	b).	List any four operations of an operating system.	1	2	2
	c).	List Multi-threading Models	2	2	2
	d).	Define turnaround time and waiting time in CPU scheduling.	2	2	2
	e).	Define Monitor?	3	2	2
	f).	Define race condition?	3	2	2
	g).	List the Contiguous memory allocation techniques	4	2	2
	h).	Define Page fault?	4	2	2
	i).	List any two file-system operations,	5	2	2
	j).	Define Access Matrix	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Explain computing system environments.	1	2	5
	b).	Interpret the types of system calls.	1	2	5
		OR			
3.	a).	Describe the major services provided by Operating System	1	2	5
	b).	Draw the Micro Kernel Operating System Structure and explain the components.	1	3	5
		UNIT-2			
4.	a).	Illustrate multi-threading models.	2	2	5
	b).	Consider the given table below and find Completion time (CT), Turn-around time (TAT), Waiting time (WT), Response time (RT),	2	3	5

		Average Turn-around time and Average Waiting time using the FCFS scheduling algorithm.																					
		<table><tr><td>PID</td><td>Arrival Time</td><td>Burst Time</td></tr><tr><td>P1</td><td>2</td><td>2</td></tr><tr><td>P2</td><td>5</td><td>6</td></tr><tr><td>P3</td><td>0</td><td>4</td></tr><tr><td>P4</td><td>0</td><td>7</td></tr><tr><td>P5</td><td>7</td><td>4</td></tr></table>	PID	Arrival Time	Burst Time	P1	2	2	P2	5	6	P3	0	4	P4	0	7	P5	7	4			
PID	Arrival Time	Burst Time																					
P1	2	2																					
P2	5	6																					
P3	0	4																					
P4	0	7																					
P5	7	4																					
		OR																					
5.	a).	Assume you have the following jobs to execute with one processor. Calculate average turnaround time and average waiting time using non pre-emptive SJF scheduling algorithm. <table><tr><td>Process</td><td>Arrival Time</td><td>Burst time</td></tr><tr><td>P1</td><td>0</td><td>6</td></tr><tr><td>P2</td><td>4</td><td>10</td></tr><tr><td>P3</td><td>4</td><td>4</td></tr><tr><td>P4</td><td>8</td><td>3</td></tr></table>	Process	Arrival Time	Burst time	P1	0	6	P2	4	10	P3	4	4	P4	8	3	2	3	5			
Process	Arrival Time	Burst time																					
P1	0	6																					
P2	4	10																					
P3	4	4																					
P4	8	3																					
	b).	Illustrate different states of a process.	2	2	5																		
		UNIT-3																					
6.	a).	Illustrate the Critical Section Problem solution for the Dining Philosophers problem using Semaphores	3	2	5																		
	b).	Describe in detail Sleep and Wakeup	3	2	5																		
		OR																					
7.		Describe Dead lock avoidance using Banker's Algorithm with an example.	3	2	10																		
		UNIT-4																					
8.	a).	Explain non contiguous memory allocation.	4	2	5																		

	b).	Find out the seek distance of read-write head using SCAN disk scheduling algorithm, when the order of cylinder request is 82, 170, 43, 140, 24, 16, 190 and current position of Read/Write head is: 50.	4	3	5
		OR			
9.	a).	Consider the following page reference string: 1 2 3 4 2 1 5 6 2 1 2 3 7 6 3 2 1 2 3 6. Find out the Number of Page faults using LRU page replacement algorithm with 3 frames.	4	3	5
	b).	Consider a disk with 200 cylinders numbered from 0 to 199. The current head position is at cylinder 70, and the requests are for cylinders 82, 170, 43, 140, 34, and 190. Assume that the head is moving towards the right direction. Find total head movement using the Look disk scheduling algorithm.	4	3	5
		UNIT-5			
10.	a).	What is the Access Matrix? Explain copy, owner and control rights with an example access matrix.	5	2	5
	b).	Explain various file access methods with suitable examples.	5	2	5
		OR			
11.	a).	Illustrate different types of directories in OS.	5	2	5
	b).	Explain free space management in OS with examples.	5	2	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23MEOE09					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
ADDITIVE MANUFACTURING					
Offered by MECH					
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE & IT)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	Explain the principle of Additive Manufacturing.	1	2	2
	b).	List the benefits of Additive Manufacturing.	1	1	2
	c).	Explain the process parameters involved in the SLA process	2	2	2
	d).	Explain the principle of Ultrasonic Additive Manufacturing	2	2	2
	e).	Explain the advantages of the Direct Metal Laser Sintering process	3	2	2
	f).	List the post-processing techniques used in additive manufacturing	3	1	2
	g).	List the advantages of Reverse Engineering	4	1	2
	h).	List the materials available for powder-based additive manufacturing.	4	1	2
	i).	Define Rapid tooling	5	1	2
	j).	List the applications of additive manufacturing in Medical Modeling	5	1	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a).	Differentiate between Conventional and Additive Manufacturing processes	1	3	5
	a)	Describe the stages in the AM process with highlighting the advantages and limitations	1	3	5
		OR			
3.	a).	Explain the generalized AM process chain.	1	3	5
	b).	Explain the file formats used in additive manufacturing	1	3	5
		UNIT-2			
4.	a).	Illustrate the working principle of the Stereolithography process (SLA).	2	3	5
	b).	Explain how a 3D model is built in the Fused Deposition Modeling	2	3	5

		process.			
		OR			
5.	a).	Explain the limitations and applications of liquid-based AM.	2	3	5
	b).	Explain the advantages and applications of laminated object manufacturing (LOM)?	2	3	5
		UNIT-3			
6.	a).	With a neat sketch, explain the working principle of the Selective Laser Sintering process.	3	3	5
	b).	Explain the role of support material removal in the AM process	3	3	5
		OR			
7.	a).	Explain the working principle of the Electron beam melting process	3	3	5
	b).	Explain how surface quality improves in the AM process	3	3	5
		UNIT-4			
8.	a).	Explain the digitization techniques used in reverse engineering.	4	3	5
	b).	Discuss various materials available for the Liquid-Based AM process	4	3	5
		OR			
9.	a).	Explain the steps involved in reverse engineering.	4	3	5
	b).	Discuss various materials available for Solid-Based AM process	4	3	5
		UNIT-5			
10.	a).	Differentiate between the direct and indirect rapid tooling methods.	5	3	5
	b).	Explain the process of 3D keltool in brief.	5	3	5
		OR			
11.	a).	Explain the implementation of the RTV epoxy tooling process.	5	3	5
	b).	Explain the applications of additive manufacturing in the automotive industry	5	3	5

COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23MEOE10					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
INDUSTRIAL SAFETY					
Offered by MECH					
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE & IT)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a).	What is the role of management in industrial safety?	1	1	2
	b).	State the importance of safety in industries	1	2	2
	c).	What are the main causes of industrial accidents?	2	1	2
	d).	What is industrial psychology in accident prevention?	2	2	2
	e).	State any two features of the Factories Act	3	2	2
	f).	What is the Explosives Act?	3	1	2
	g).	What is meant by fire prevention?	4	1	2
	h).	What is meant by first aid for burns?	4	1	2
	i).	Define egress in building safety.	5	2	2
	j).	Define fire load	5	2	2
5 x 10 = 50 Marks					
		UNIT-1			
2.		Determine the role of management in implementing industrial safety measures in an organization.	1	3	10
		OR			
3.		Use the provisions of the Factories Act to ensure worker safety in industries	1	3	10
		UNIT-2			
4.	a).	Explain how education and training programs improve safety awareness among workers.	2	2	5
	b).	Demonstrate the importance of housekeeping in accident prevention.	2	3	5

		OR			
5.	a).	Describe the role of first aid in minimizing the impact of industrial accidents.	2	2	5
	b).	Explain about industrial psychology in reducing workplace accidents.	2	2	5
		UNIT-3			
6.	a).	Determine how engineering controls can reduce chemical hazards in a manufacturing plant.	3	3	5
	b).	Demonstrate how industrial hygiene practices help prevent occupational diseases.	3	3	5
		OR			
7.	a).	Determine the health risks associated with excessive industrial noise.	3	3	5
	b).	Explain the relationship between health, safety, and the physical work environment.	3	2	5
		UNIT-4			
8.	a).	Apply the concept of the fire triangle to explain how fires can be prevented in industrial workplaces.	4	3	5
	b).	Predict the risks associated with improper maintenance of fire trucks and fire safety equipment.	4	3	5
		OR			
9.	a).	Apply suitable active and passive fire protection systems to improve safety in a factory building.	4	3	5
	b).	Demonstrate the first aid treatment for burn injuries caused by fire accidents	4	3	5
		UNIT-5			
10.		Demonstrate how fire-resistant materials help prevent the spread of fire.	5	3	10
		OR			
11.		Determine the fire safety requirements necessary for high-rise buildings.	5	3	10

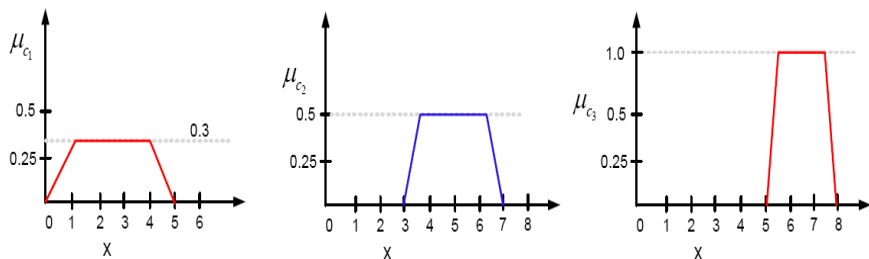
COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks

Course Code: B23BSOE04					
SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (A)					R23
IV B.Tech. I Semester MODEL QUESTION PAPER					
FUZZY SETS AND FUZZY LOGIC					
Offered by E&H					
(Offered to AIDS, AIML, CE, CIC, CSBS, CSE, CSG, CSIT, ECE, EEE, IT & ME)					
Time: 3 Hrs.			Max. Marks: 70 M		
Answer Question No.1 compulsorily					
Answer ONE Question from EACH UNIT					
Assume suitable data if necessary					
10 x 2 = 20 Marks					
			CO	KL	M
1.	a)	Define a fuzzy set.	1	2	2
	b)	Define Membership function.	1	2	2
	c)	Define fuzzy relation.	2	2	2
	d)	Write any two properties on fuzzy relations.	2	3	2
	e)	What is fuzzification?	3	3	2
	f)	What is defuzzification?	3	3	2
	g)	Write any two logical operations in fuzzy logic.	4	3	2
	h)	Write the If-then rule in fuzzy logic.	4	3	2
	i)	Write any one application on fuzzy logic controller.	5	3	2
	j)	Write any one fuzzy rule used in Air conditioner.	5	3	2
5 x 10 = 50 Marks					
		UNIT-1			
2.	a)	Compare Fuzzy set and Crisp set with suitable example.	1	2	5
	b)	List the Fuzzy operations with examples.	1	3	5
		OR			
3.		We want to compare two sensors based upon their detection levels and gains setting for a universe of discourse of gain settings $X = \{0, 20, 40, 60, 80, 100\}$, the sensor detection levels for the monitoring of a standard item provides typical membership functions to represent the detection levels for each of the sensors; these are given below in standard discrete form $\tilde{S}_1 = \left\{ \frac{0}{0} + \frac{0.5}{20} + \frac{0.65}{40} + \frac{0.85}{60} + \frac{1}{80} + \frac{1}{100} \right\}$, $\tilde{S}_2 = \left\{ \frac{0}{0} + \frac{0.45}{20} + \frac{0.6}{40} + \frac{0.8}{60} + \frac{0.95}{80} + \frac{1}{100} \right\}$ Find the following using standard fuzzy operations	1	3	10

		(i) $\tilde{S}_1 \cup \tilde{S}_2$ (ii) $\tilde{S}_1 \cap \tilde{S}_2$ (iii) $\tilde{S}_1^c \cup \tilde{S}_2$ (iv) $\tilde{S}_1^c \cap \tilde{S}_2$ (v) $\tilde{S}_1 - \tilde{S}_2$			
		UNIT-2			
4.		In the field of computer networking there is an imprecise relationship between the level of use of a network communication bandwidth and the latency experienced in peer-to-peer communications. Let X be the fuzzy set of use levels (in terms of percentage of full bandwidth used) and Y be a fuzzy set of latencies (in milliseconds) with the following membership functions: $X = \left\{ \frac{0.2}{10} + \frac{0.5}{20} + \frac{0.8}{40} + \frac{1}{60} + \frac{0.6}{80} + \frac{0.1}{100} \right\} \quad Y = \left\{ \frac{0.3}{0.5} + \frac{0.6}{1} + \frac{0.9}{1.5} + \frac{1}{4} + \frac{0.6}{8} + \frac{0.3}{20} \right\}$ Find the Cartesian product represented by $R = X \times Y$. Now, suppose we have a second fuzzy set of bandwidth usage given by $Z = \left\{ \frac{0.3}{10} + \frac{0.6}{20} + \frac{0.7}{40} + \frac{0.9}{60} + \frac{1}{80} + \frac{0.5}{100} \right\}$ Find $S = Z \circ R$ using max-min composition.	2	3	10
		OR			
5.	a)	What is the composition of fuzzy relations? Illustrate with example.	2	3	5
	b)	Let R be a fuzzy relation defined on the set of cities representing the concept 'very near'. Discuss the properties of fuzzy equivalence relation.	2	3	5
		UNIT-3			
6.	a)	The following are the fuzzified outputs of three fired rules:  Find the crisp output from the union of fuzzified outputs by using any one of the defuzzification methods.	3	3	5
	b)	Suppose a fuzzy set young is defined as follows $\text{young} = \{(15,0.5), (20,0.8), (25,0.8), (30,0.5), (35,0.3)\}$. Find the crisp value of young using mean of maxima method	3	3	5
		OR			
7.	a)	Explain centroid method for defuzzification.	3	3	5

	b)	The width of a road as narrow and wide is defined by two fuzzy sets, whose membership functions are plotted as shown below. If a road with its degree of membership value is 0.4 then what will be its width (in crisp) measure.	3	3	5
		UNIT-4			
8.	a)	Construct a truth table for $(p \rightarrow r) \wedge (q \rightarrow r) \vee (\sim p \wedge q)$	4	3	5
	b)	Write the symbolic form of the following statement "you cannot ride the roller coaster if you are under 4 feet tall unless you are older than 16 years old"	4	3	5
		OR			
9.		Given P : Bob is brilliant ; $T(P) = 0.8$; Q : John is brilliant; $T(Q) = 0.6$; Find the truth values of the following fuzzy propositions (i) Bob is not brilliant (ii) Bob is brilliant and so is John (iii) Either Bob or John is brilliant (iv) if Bob is brilliant then so is John (v) Bob is brilliant if and only if John is brilliant	4	3	10
		UNIT-5			
10.		Explain in detail any one application of Fuzzy Logic Controller.	5	3	10
		OR			
11		Explain how would you design a washing machine to have fuzzy inference in a real world environment. In your discussion include the following factors (i) A practical choice of the fuzzy input & output variables (ii) A practical no. of quantizations and choice of linguistic labels (iii) Draw and indicate a reasonable membership functions for each fuzzy variable. (iv) Indicate practical values of universe of discourse (v) Three examples of rules	5	3	10

		(vi) The most practical approach of defuzzification technique			
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COURSE OUTCOME

KL-KNOWLEDGE LEVEL

M-MARKS

NOTE: Questions can be given as A,B splits or as a single Question for 10 marks



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